

Paying With Promises

Dynamic Contracting with Ex Post Participation

Mitchell Watt

Monash University

October 6, 2026

Monash Brown Bag

Some days, our jobs suck.

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Mitch Watt <mitch.watt@monash.edu>

Mandatory Training

1 message

Vicki Briard <vicki.briard@monash.edu>
Bcc: Mitch.Watt@monash.edu

15 August 2025 at 13:42

Good afternoon

Our records indicate that you have not yet completed the **"Ergonomic Principles"** module. This is a mandatory requirement for all staff.

Please complete the module as soon as possible via the **MyDevelopment** portal. You can locate it by navigating to:

Staff Portal → HR & Finance → myDevelopment

Search for Ergonomic Principles using the Search icon in the top RH corner

The module should take approximately 20 minutes to complete.

Thank you for your attention to this matter. If you have any questions or need assistance accessing the module, feel free to reach out.

Kind regards

Vicki

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MONASH University

[< Expense Claims](#)

May Expense Claims AUD 1,107.51

Finalised for Accounting/Payment

Claim number: **MHK6M6**

[Claim Details](#) ▾

[Print/Share](#) ▾

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Expenses (6)

View:

Detail ▾

Date ⚙	Receipt ⬆⬆	Attributes	Expense Type ⬆⬆	Transaction Date ⬆⬆
20/05/2026		 	Food & Drink Travel - Australia	20/05/2026
15/04/2026		 	Professional Memberships - Overseas	15/04/2026
15/04/2026		 	General Expenses - Overseas	15/04/2026

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Australian Government

Australian Research Council

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Yet, our wages are fixed on an annual basis.

ENTERPRISE AGREEMENTS

Staff Salary Rates

Home | Staff Salary Rates | Academic Staff Annual Salary Rates

Academic Staff Annual Salary Rates

Professorial and Clinical Loadings - Academic Staff

Sessional Rates Including Casual Academic Research Assistant Rates

Professional (and Trades) Staff Annual Salary Rates

Casual Professional Staff Salary Rates

Academic Staff Annual Salary

Academic Level	Salary Step	3% As From 3/12/22	4% As From 3/6/23
A	8	\$75,115	\$78,120
	7	\$79,413	\$82,590
	6	\$83,703	\$87,051
	5	\$87,996	\$91,516

Other Employment Relationships

Many government jobs have fixed wage schedules, with residual heterogeneity in assignments.

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- ▶ **Teachers:** placements to remote schools are harder to fill than city schools.
- ▶ **Diplomats:** Foreign Service posts differ in hardship and living conditions.
- ▶ **Military personnel:** in the navy, sea tours and shore duty involve different demands and time away.

Beyond Employment

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- ▶ **Legal Aid NSW:** fee scales precede individual cases; panel lawyers can accept or decline each offer.
- ▶ **Facilities maintenance:** a framework contract sets rates before work at each site is specified.
- ▶ **Uber drivers:** fare rules leave some trip costs unpriced, including time spent in an airport queue.
- ▶ **Truckload freight:** route rates are agreed before specific loads are tendered to carriers.

Compensation Through Future Assignments

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- ▶ **Queensland teachers:** remote service earns transfer points and priority for preferred locations.
- ▶ **U.S. Navy:** the Sea Shore Flow policy pairs prescribed sea tours with shore duty.
- ▶ **Automotive suppliers:** past deliveries give preferred status and access to new business.
- ▶ **Uber drivers:** drivers accepting a short airport trip can preserve queue position when they return.

This Paper

I study a principal who assigns projects to agents under **fixed payment terms**.

- ▶ Each period, up to K projects can be assigned among $N \geq K$ agents.
- ▶ Project costs are realised and publicly observed.
- ▶ The principal benefits from completed projects and can commit to future assignments.
- ▶ An agent sees her assignment and promised future treatment, then may accept, refuse, or quit.

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Question. How should the principal use future assignments to induce agents to accept costly projects?

Related Work

- ▶ **Payment terms and future business in practice.** Harris and Nguyen (2025); Blouin and Macchiavello (2019); Macchiavello and Morjaria (2015); Bernstein (2015); Chen et al. (2024).
- ▶ **Repeated projects and relational contracts.** Forand and Zápal (2020); Samuelson and Stacchetti (2017); Andrews and Barron (2016).
- ▶ **Assignment as an incentive.** Castro et al. (2021); Ji and Vong (2025); Eliaz et al. (2025).
- ▶ **Future allocation rights.** Bird and Frug (2019); Guo and Hörner (2018); Combe et al. (2025).
- ▶ **Recursive contracts.** Spear and Srivastava (1987); Abreu et al. (1990).

This paper: commitment to general, history-dependent assignment mechanisms; ex post participation after the realised assignment; potentially different discount factors for the principal and agents.

Roadmap

1. Model
2. Single-agent benchmark
3. Optimal assignment rule
4. Long-Run Dynamics
5. Multiple agents and projects
6. Applications

Model

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After observing the task and promised continuation, the worker accepts or rejects the assignment.

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4. Accepted tasks are completed; rejected or unassigned tasks disappear.

Payoffs

Let $x_t \in \{0, 1\}$ denote assignment and $y_t \in \{0, 1\}$ denote acceptance.

- ▶ Completion is $x_t y_t$.
- ▶ The firm receives **flow payoff** $x_t y_t$.
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- ▶ The firm receives **flow payoff** $x_t y_t$.
- ▶ The worker receives **flow payoff** $v_t x_t y_t$.
- ▶ **Discount factors** may differ: δ_P for the firm and δ_A for the worker.

The firm maximises $(1 - \delta_P) \sum_{t \geq 0} \delta_P^t x_t y_t$, while the worker evaluates $(1 - \delta_A) \sum_{t \geq 0} \delta_A^t v_t x_t y_t$.

Contracts

A contract is a **history-dependent** assignment rule.

At history h_t , the firm chooses **assignment probability** $X_t(v_t, h_t)$.

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I impose:

- ▶ **public histories**: task values, assignments, and acceptances are observed;
- ▶ **commitment**: the firm can commit to future assignment rules;
- ▶ **tie-breaking**: an indifferent worker accepts.

The constraint of interest is ex post participation after each task realisation.

Recursive Formulation

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Notes:

- (a) It is without loss to restrict to **deterministic** contracts.
- (b) An optimal rule can be represented with every assigned task accepted; an **ex post participation constraint** makes acceptance optimal.
- (c) **Off-path**, the principal punishes the agent with the lowest feasible continuation promise (0).

Bellman Problem

The firm solves:

$$\Phi(u) = \max_{\substack{X(\cdot; u): \mathcal{V} \rightarrow \{0,1\} \\ u'(\cdot; u): \mathcal{V} \rightarrow \mathcal{U}}} \mathbb{E} \left[(1 - \delta_p) X(v; u) + \delta_p \Phi(u'(v; u)) \right],$$

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subject to a **promise-keeping constraint**:

$$\mathbb{E} \left[(1 - \delta_A) v X(v; u) + \delta_A u'(v; u) \right] \geq u \quad (\text{PK} - \lambda(u))$$

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and an **ex post participation constraint**:

$$u'(v; u) \geq \underline{u}(v) X(v; u) \quad \text{for every } v, \quad \underline{u}(v) = \left[-\frac{1 - \delta_A}{\delta_A} v \right]_+ \quad (\text{PC} - \mu(v; u))$$

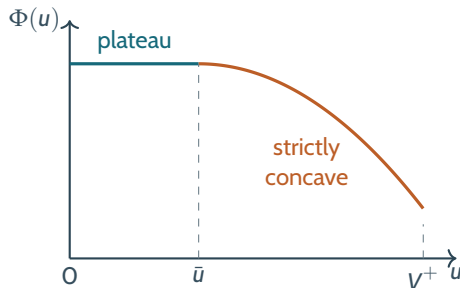
Value Function Properties

Proposition 1

The Bellman problem has a unique fixed point $\Phi : \mathcal{U} \rightarrow [0, 1]$. It is nonincreasing, concave, and continuous.

- ▶ There is a **plateau** $[0, \bar{u}]$: Φ is constant there.
- ▶ Above $\bar{u}$, promises are costly: Φ is **strictly decreasing** and **strictly concave** on $(\bar{u}, V^+]$.
- ▶ Above the plateau, the marginal cost of a larger promise is positive and rises with u .

Intuition for plateau: at $u = 0$, firm wants to assign positive-value tasks $\rightsquigarrow$ strictly positive utility (any negative value tasks assigned must be compensated by promises and contribute ≥ 0 to promised utility).



▶ Strict concavity

Optimal Single-Agent Contract

Static Benchmark

A **static** contract uses the same $X : \mathcal{V} \rightarrow \{0, 1\}$ after every history. Since each completed task gives the firm payoff 1, optimal static assignments are **cutoffs**: $X(v) = 1\{v \geq c\}$.

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For cutoff $c < 0$, the worker's payoff from accepting the task at the cutoff is

$$H(c) = \underbrace{\delta_A \mathbb{E}[v 1\{v \geq c\}]}_{\text{discounted value of future assignments}} + \underbrace{(1 - \delta_A)c}_{\text{current payoff at the cutoff}}.$$

Static cutoff c is **feasible** when $H(c) \geq 0$: future assignments compensate for taking the cutoff task.

Proposition 2: Optimal Static Contract

The **optimal static contract** is a cutoff rule $X^S(v) = 1\{v \geq c^S\}$, where

$$c^S = \begin{cases} \underline{v}, & \text{if } H(\underline{v}) \geq 0, \\ \text{the unique root of } H(c) = 0 \text{ in } (\underline{v}, 0), & \text{if } H(\underline{v}) < 0. \end{cases}$$

It delivers $U^S = \mathbb{E}[v 1\{v \geq c^S\}]$ and $\Pi^S = \Pr(v \geq c^S)$.

When is the static contract dynamically optimal?

When $\mathbb{E}[v] \geq \frac{1-\delta_A}{\delta_A}(-\underline{v})$, the firm can assign **every task** with the fixed promise $\mathbb{E}[v] \rightsquigarrow$ optimal.

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Otherwise, dynamic contracts do **strictly better**:

Proposition 4: Static Full Assignment or Dynamic Improvement

Either:

1. the optimal static contract has cutoff $c^S = \underline{v}$, so it assigns every task; or
2. the optimal static contract has cutoff $c^S > \underline{v}$, and the optimal dynamic contract strictly improves on the optimal static contract: $\Phi(0) > \Pi^S$.

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- ▶ A task just below the static cutoff needs only a little extra compensation, but completing it gives the principal a whole extra task.

Henceforth, assume $\mathbb{E}[v] < \frac{1-\delta_A}{\delta_A}(-\underline{v})$, so full assignment is statically infeasible.

▶ Proof sketch

Characterising the Optimal Dynamic Contract

At state u , the firm chooses $X : \mathcal{V} \rightarrow \{0, 1\}$ and $u' : \mathcal{V} \rightarrow \mathcal{U}$ to solve

$$\Phi(u) = \max_{X, u'} \mathbb{E} \left[(1 - \delta_P) X(v) + \delta_P \Phi(u'(v)) \right],$$

subject to

$$(\text{PK} - \lambda(u)) \quad \mathbb{E}[(1 - \delta_A)vX(v) + \delta_A u'(v)] \geq u, \quad (\text{PC} - \mu(v; u)) \quad u'(v) \geq X(v)\underline{u}(v) \quad \forall v.$$

1. **Solve for u' given X .** Fix a candidate assignment rule and solve the continuation-promise problem.
2. **Given u' , show optimal X is cutoff.** After substituting optimal continuation promises, show assignment is monotone in v .
3. **Solve jointly for X and u' .** Combine the two steps to characterise the optimal contract.

Solving for u' given X

Fix $u \in (\bar{u}, V^+)$ and an assignment rule X . Apart from participation, v enters only through **average current utility** $\mathbb{E}[vX(v)]$:

$$\underbrace{\delta_A \mathbb{E}[u'(v)]}_{\text{future utility contribution}} \geq \underbrace{u}_{\text{utility owed}} - \underbrace{(1 - \delta_A) \mathbb{E}[vX(v)]}_{\text{current utility contribution}}.$$

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Choose a **common target** t_λ , unless participation requires more:

$$u'_X(v; \lambda) = \max\{t_\lambda, \ell_X(v)\}, \quad \ell_X(v) = X(v; u)\underline{u}(v).$$

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- ▶ $\ell_X(v) \leq t_\lambda$: the same promise $u' = t_\lambda$ across these states.
- ▶ $\ell_X(v) > t_\lambda$: participation binds; promise just enough for acceptance.

Optimal Assignments are Cutoff Rules

Theorem 1(i): Assignment

There is a cutoff $\gamma : [0, V^+] \rightarrow [\underline{v}, 0]$ such that $X(v; u) = \mathbf{1}\{v \geq \gamma(u)\}$. The cutoff is constant on $[0, \bar{u}]$, nondecreasing above $\bar{u}$, and $\gamma(V^+) = 0$.

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Why a cutoff? Fix the marginal promise cost λ . Compared with a lower-value task, a higher v gives:

1. the **same payoff to the firm** from completion;
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Why does the cutoff rise with u ? The cutoff balances the payoff from completion against the **marginal cost of additional promises**, captured by λ . Above the plateau, higher u raises λ , making future obligations more costly.

Putting These Together

Assignment:

$$X(v; u) = \mathbf{1}\{v \geq \gamma(u)\}.$$

Continuation ($\bar{u} < u < V^+$):

$$u'(v; u) = \begin{cases} t(u), & v < \gamma(u), \\ \max\{\underline{u}(v), t(u)\}, & v \geq \gamma(u). \end{cases}$$

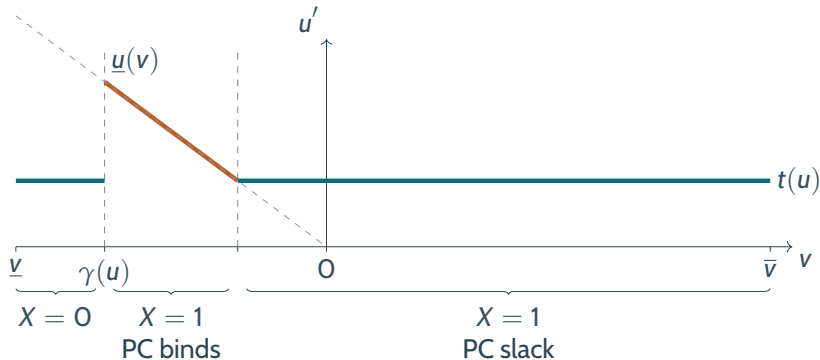
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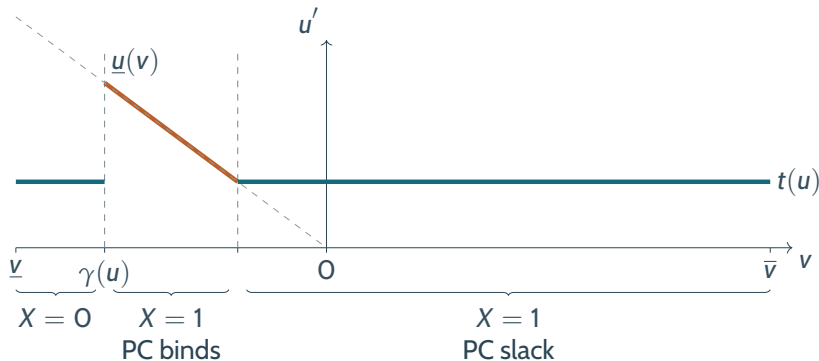
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Assignment:

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What determines the cutoff $\gamma(u)$, or equivalently the slack target $t(u)$?

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Above the plateau, promise keeping binds: once $t(u)$ is known, it pins down $\gamma(u)$. Let $\lambda \in \Lambda(u)$ and $q = \delta_A / \delta_P$.

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The slack target maximises the continuation terms of the Lagrangian:

$$t(u) \in \arg \max_{z \in [0, V^+]} \{ \delta_P \Phi(z) + \delta_A \lambda z \}.$$

Putting These Together

Above the plateau, promise keeping binds: once $t(u)$ is known, it pins down $\gamma(u)$. Let $\lambda \in \Lambda(u)$ and $q = \delta_A / \delta_P$.
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More generally, we have the **Euler equation**:

$$\lambda_{t+1} = q\lambda_t + \frac{\mu_t}{\delta_P} \in \Lambda(u_{t+1}), \quad \mu_t \geq 0,$$

where μ_t is the multiplier on the participation constraint.

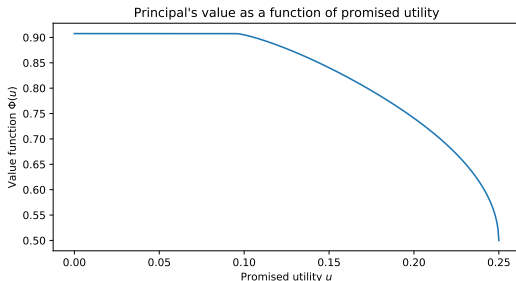
► Derivation

Long-Run Dynamics

Agent More Patient: $q > 1$

Proposition 5(i) - Patient Agent

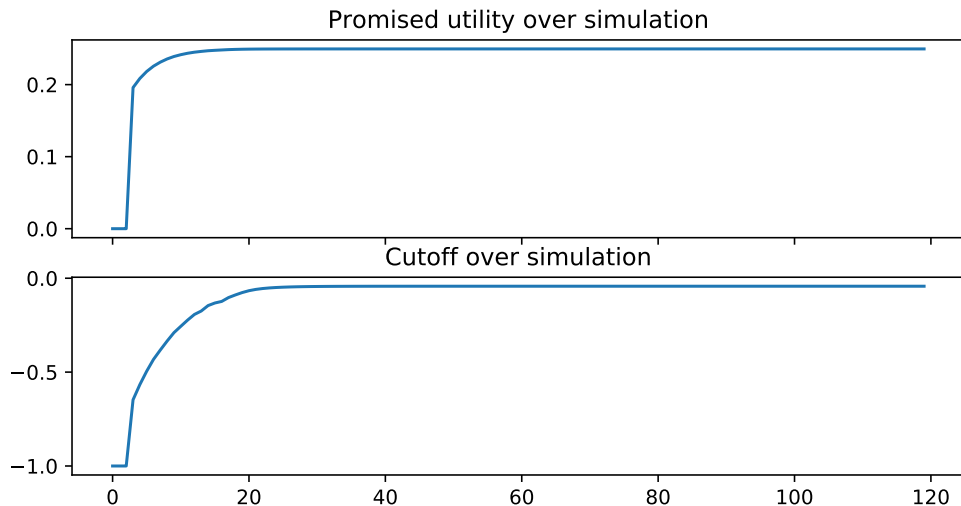
If $q > 1$, promised utility eventually rises toward its feasible maximum V^+ . The assignment cutoff approaches zero, making negative-value tasks increasingly rare.



- ▶ Promised utility is relatively cheap for the principal, so the contract uses **backloading**: compensation comes through future task access.
- ▶ The costate equation is $\lambda_{t+1} = q\lambda_t + \mu_t/\delta_P$, so positive costates rise when $q > 1$.
- ▶ Strict concavity maps higher costates into higher promises, pushing u_t toward V^+ .
- ▶ The cutoff approaches zero, making negative-value assignments increasingly rare.

Example Time Paths: $q > 1$

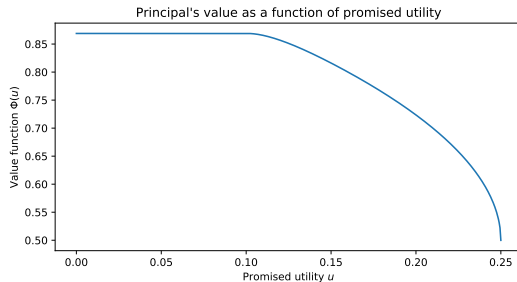
$$v \sim \text{Unif}[-1, 1], \delta_A = 0.8 > \delta_P = 0.7$$



Equal Patience: $q = 1$

Proposition 5(ii) - Equal Patience

If $q = 1$, promised utility converges to an interior level determined by the history. The cutoff approaches a fixed negative threshold, so mildly negative-value tasks remain eligible for assignment.

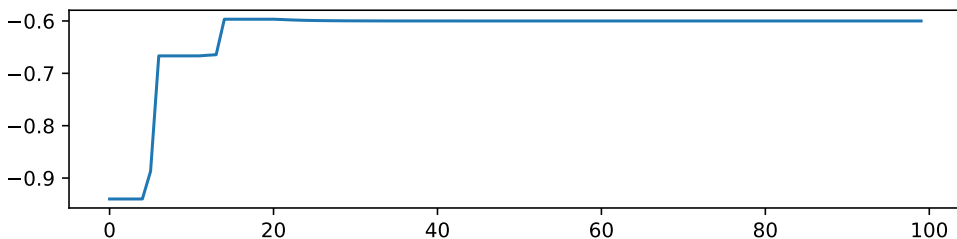
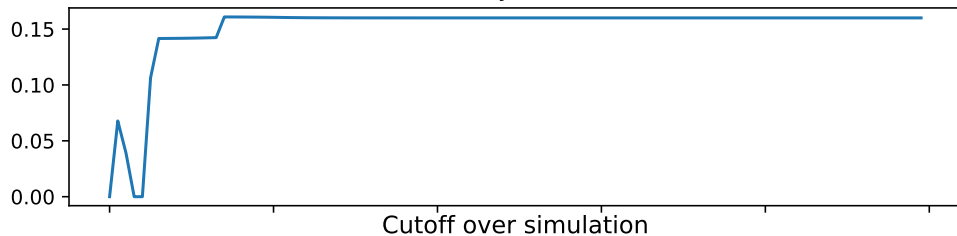


- ▶ Future compensation has the same discount cost for both sides, so slack periods leave the costate unchanged.
- ▶ When participation is slack, $\lambda_{t+1} = \lambda_t$; a positive multiplier on binding participation pushes λ_t up.
- ▶ The limiting cutoff stays below zero, so mildly negative-value tasks remain eligible for assignment.

Example Time Paths: $q = 1$

$$v \sim \text{Unif}[-1, 1], \delta_A = \delta_P = 0.8$$

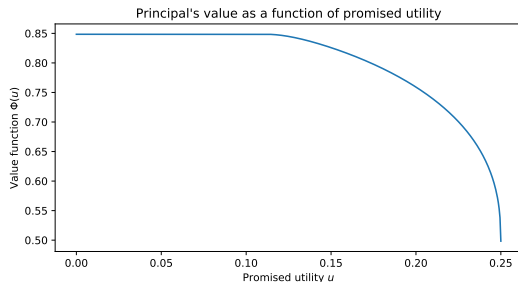
Promised utility over simulation



Principal More Patient: $q < 1$

Proposition 5(iii) - Patient Principal

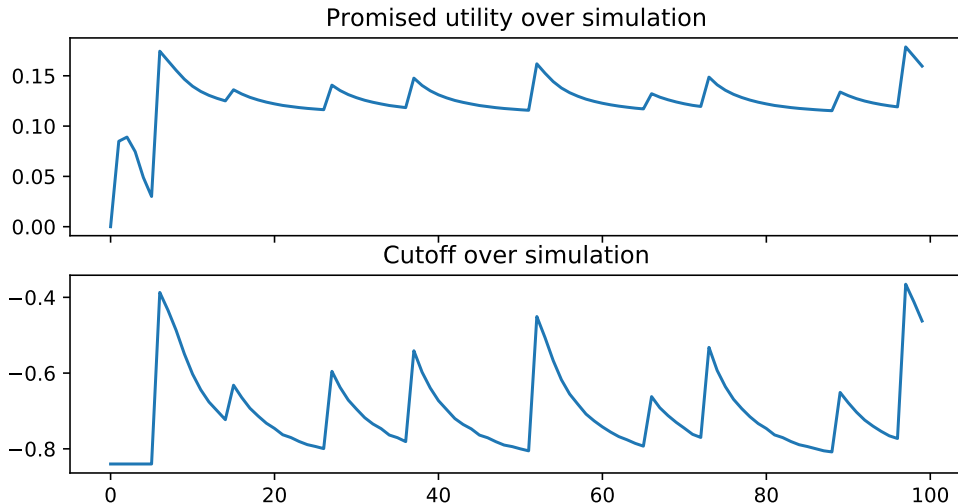
If $q < 1$, promised utility leaves the plateau in finite time and then keeps fluctuating. It falls toward the plateau when participation is slack and rises after some difficult assignments.



- ▶ Promised utility is relatively expensive for the principal, so the contract runs down promises when participation is slack.
- ▶ Then $\lambda_{t+1} = q\lambda_t < \lambda_t$, moving the target promise back toward the plateau.
- ▶ Some unpleasant assignments raise the promise again, generating persistent cycles of effort and reward.
- ▶ Positive- and negative-value tasks are assigned forever along these cycles.

Example Time Paths: $q < 1$

$$v \sim \text{Unif}[-1, 1], \delta_A = 0.8 < \delta_P = 0.9$$



Optimal Multi-Agent Contract

From One Worker to Many

The only decision in the single-agent case is whether or not to assign the realised task. The multi-agent case adds the decision of who receives each assigned task.

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Objects

- ▶ $N \geq 2$ workers and $K \leq N$ tasks arrive each period.
- ▶ Workers share the same discount factor δ_A .
- ▶ Task values are $v = (v_1, \dots, v_K) \sim F^K$.
- ▶ The state is a promise vector $u = (u_1, \dots, u_N)$.

Choices and Constraints

- ▶ Assignment is a matching matrix $X(v; u) \in \{0, 1\}^{N \times K}$.
- ▶ Each task goes to at most one worker; each worker gets at most one task.
- ▶ Promise keeping and ex post participation hold worker by worker.

Feasible Promise Polytope

m workers can receive at most the best $\min\{m, K\}$ positive tasks per period. This bounds their total promise.

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$$\bar{V}_m = \mathbb{E} \left[\sum_{r=K-m+1}^K [v_{(r)}]_+ \right], \quad \bar{V}_0 = 0.$$

For $m > K$, set $\bar{V}_m = \bar{V}_K$.

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$$\mathcal{U}^{N,K} = \left\{ u \in \mathbb{R}_+^N : \sum_{i \in S} u_i \leq \bar{V}_{|S|} \text{ for every } S \subseteq \{1, \dots, N\} \right\}.$$

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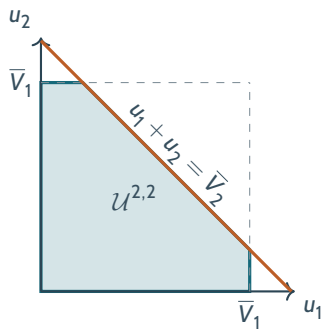
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A symmetric, downward-closed polytope.



Schur-Concavity and Equalisation

$\Phi^{N,K}$: continuous, nonincreasing, concave; a plateau. Worker **symmetry** adds a preference for equalisation.

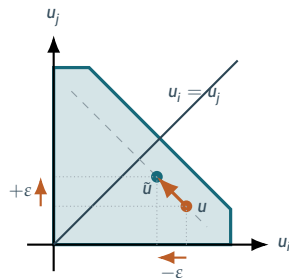
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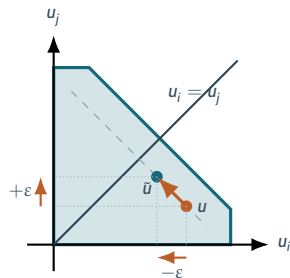
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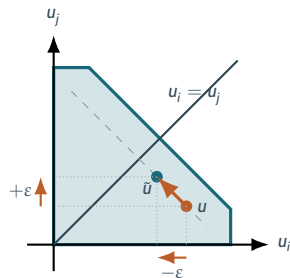
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⇒ Same total promise: more **equal** is weakly better for the principal.



► Schur-concavity

Optimal Assignment: Theorem and Intuition

Let $Y_i = \sum_k v_k X_{ik}$ be worker i 's current task value (0 if unassigned).

Theorem 2: Assortative Allocation on Average

There is an optimal rule with

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All positive tasks assigned; negative tasks assigned least costly first.

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Thus current utility enters as $(1 - \delta_A)\mathbb{E}[\sum_i \lambda_i Y_i]$. **Higher-costate workers receive more weight:** utility delivered today saves more costly future compensation.

Why Priority is Sometimes Soft

Distinct costates: sort tasks by marginal promise cost. If $\lambda_j > \lambda_i$ and $y > x$, giving y to j rather than i raises the current-utility term by

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The firm's completion payoff is unchanged. Participation may require an equalising continuation transfer; its costate-weighted loss cannot exceed this gain, and continuation value weakly rises.

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- ▶ $\lambda = 0$ supports plateau states; promises that are close to equal can also lead to tied costates $\lambda_i = \lambda_j$.
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Always giving j the better task may create too large a current gap. **Occasional reversals** keep assignments consistent with the promised gap; priority holds **on average**.

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- ▶ **The principal maximises its own payoff.** Its objective values completed tasks. Worker symmetry and the cost of fulfilling promises generate the force toward equalisation.

Long-Run Dynamics

Theorem 3: Long-Run Assignments

When a static rule cannot assign every task, relative patience determines which tasks continue to be assigned:

- ▶ **Agent more patient ($q > 1$):** negative-value tasks are assigned only finitely often.
- ▶ **Equal patience ($q = 1$):** positive- and negative-value tasks recur. Matching choices approach a stationary benchmark selected by the history.
- ▶ **Principal more patient ($q < 1$):** positive- and negative-value tasks recur. For an optimal rule, the worker taking the worst task regularly moves from the lowest to the highest priority group in the next period.

Takeaway. Future assignments compensate difficult tasks in the long run when the principal is at least as patient as the workers. With a more patient principal, priority reversals recur.

The Value of Flexible Assignment

Compare K separate relationships, each tied to one project stream, with a common pool of K agents and K projects per period. All relationships start with zero promises.

Pooling creates value

With at least two project streams, pooling strictly raises the principal's payoff whenever a static rule cannot assign every project.

- ▶ An attractive project in one stream can compensate an agent for accepting a difficult project in another.
- ▶ Flexibility lets the principal direct attractive projects toward larger claims and difficult projects toward smaller claims.
- ▶ **Procurement interpretation:** pool orders across products, plants, or markets to support cooperation through future business.

The Value of Spare Capacity

Can maintaining more agents than simultaneous projects improve incentives?

- ▶ **One project per period:** extra agents create no gross value. Only the total outstanding promise matters.
- ▶ **Several projects per period:** an extra agent can strictly improve the principal's payoff. The paper establishes examples of this gain.
- ▶ **Mechanism:** spare capacity allows an agent who accepted a very difficult project to receive favorable treatment while other agents handle the remaining work.
- ▶ **Applications:** spare staffing, an additional approved supplier, or a larger pool of platform providers.

Comparative Statics: Agent Patience and Project Values

Starting from zero promises, these results apply to both one and several agents.

- ▶ **More patient agents raise the principal's payoff.** The gain is strict whenever the principal falls short of assigning all projects. Future business becomes more effective compensation for difficult work.
- ▶ **The limits are intuitive.** As $\delta_A \rightarrow 0$, only positive-value projects are assigned in the limit. As $\delta_A \rightarrow 1$, the principal's discounted assignment fraction approaches one.
- ▶ **Better project values weakly raise the principal's payoff.** If agent values improve in the sense of first-order stochastic dominance, the old assignment process remains feasible.

Comparative Statics: Principal Patience

The principal's normalized payoff P measures the discounted number of assignments per period.

- ▶ **Greater principal patience weakly lowers P .** The decline is strict when static full assignment is infeasible. More weight falls on later periods, when earlier promises must be fulfilled.
- ▶ **Total discounted assignments increase.** The unnormalized sum, $P/(1 - \delta_P)$, strictly rises with principal patience.
- ▶ **The gain over the static benchmark shrinks.** A static rule delivers the same expected assignments each period; its normalized payoff is independent of principal patience.

Connection to long-run dynamics. When static full assignment is infeasible, difficult assignments recur once the principal is at least as patient as the agents.

Discussion

Motivating Example

Allocation of School Teachers



- ▶ Teachers are allocated centrally.
- ▶ Jobs appear over time: teacher retirements, transfers, and new demand.
- ▶ Assignment values vary with locational preferences and school difficulty.
- ▶ Public-sector pay scales limit salary adjustment across postings.
- ▶ **Transfer points:** teachers accrue priority while matched to less desirable schools.

▶ [More details](#)

Queensland Teacher Transfers

	Year 1	Year 2	Year 3	Year 4	Year 5
Rating 7	11	11	17	20	22
Rating 6	7	7	7	10	12
Rating 5	5	5	5	7	8
Rating 4	4	4	4	6	7
Rating 3	3	3	3	4	5
Rating 2	2	2	2	2	2
Rating 1	1	1	1	1	1

Transfer points by school rating and years of service.

Institution.

- ▶ Teachers are allocated through a centralised transfer system.
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Interpretation. Hardship service builds standing for future transfers.

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- ▶ **Kenyan roses.** Supply disruptions force exporters to prioritize orders; deliveries depend on the value of established relationships. [Macchiavello and Morjaria \(2015\)](#).

▶ [Skip to conclusion](#)

Other Assignment Currencies

Queue priority

- ▶ Uber, Lyft, and DiDi use airport or short-trip queue rules.
- ▶ A bad trip preserves or upgrades future access to better trips.

Point systems

- ▶ Amazon Flex drivers accumulate standing through completed blocks.
- ▶ Higher standing unlocks preferred stations, times, and blocks.

Rotation rules

- ▶ Foreign service and naval careers link hardship tours to later desirable postings.
- ▶ Residency programmes use point bidding for preferred rotations.

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- ▶ **Hiring or qualifying another supplier.** With a maintenance cost c per relationship, initial net value is $\Phi^{N,K}(0) - Nc$. An extra relationship is worthwhile when its gross benefit exceeds c .
- ▶ **Spare capacity has an incentive benefit.** With several projects, it can preserve output while earned priority is honored. This benefit disappears once the existing pool supports full assignment every period.

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Hiring and Termination: Implications

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Further analysis: optimal turnover requires hiring and termination costs, and a rule for settling promises through severance, buyouts, or future business.

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- ▶ **The continuation logic carries over.** Under suitable conditions, the single-agent promise is the larger of a target and the minimum required to induce the recommended action.

Conclusion

- ▶ With payment terms fixed in advance, future projects can compensate agents who accept costly assignments today.
- ▶ With one agent, the principal assigns projects above a cutoff that rises weakly with the agent's promised utility.
- ▶ Relative patience determines whether negative-value projects gradually disappear or keep recurring.
- ▶ With several agents, an optimal rule gives agents owed more assignments worth at least as much on average. Similar promises can allow reversals in individual periods.
- ▶ **Applications:** Transfer points, preferred-supplier status, and queue priority reward current service with access to future opportunities.

Less Obvious Implications

- ▶ **Patient workers need not do difficult work forever.** When static full assignment is infeasible, negative-value assignments eventually stop if agents are more patient than the principal. They recur with positive long-run frequency if the principal is at least as patient.

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- ▶ **Priority need not be a strict queue.** An optimal rule gives workers with higher promises assignments worth at least as much on average. Tied costates can allow reversals in individual periods.

Appendix

Deriving the Costate Equation

Fix state u_t , realised task value v_t , and write $\lambda_t = -\Phi'(u_t)$. The terms in the Lagrangian that depend on the continuation promise $u_{t+1} = u'(v_t; u_t)$ are

$$\delta_P \Phi(u_{t+1}) + \delta_A \lambda_t u_{t+1} + \mu_t(v_t) \{u_{t+1} - \underline{u}(v_t) X_t(v_t)\}.$$

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- ▶ Slack participation: $\mu_t(v_t) = 0$, so the costate moves by relative patience q .
- ▶ A positive multiplier requires binding participation and raises the next costate.

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Promises are bounded and $\delta_A < 1$, so the largest gap is zero. Each positive marginal cost selects **one promise**, ruling out linear segments above the plateau.

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Proof: Strict Concavity

Let $A_\lambda = \arg \max_{u \in [0, V^+]} \{\Phi(u) + \lambda u\}$: the promises supported by marginal cost $\lambda > 0$. Define the largest gap $M = \sup_{\lambda > 0} \text{diam}(A_\lambda) \leq V^+$.

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Every positive costate selects a single promise. A linear segment above the plateau would have the same positive costate at multiple promises, so Φ is **strictly concave** there.

Proof: Proposition 4

Static full assignment gives payoff 1 every period, so it is dynamically optimal whenever it is feasible.

Suppose instead $c^S > \underline{v}$. Then $c^S \in (\underline{v}, 0)$, static feasibility binds, and $U^S = \mathbb{E}[v \mathbf{1}\{v \geq c^S\}] = \underline{u}(c^S) < V^+$.

1. Pick small $\varepsilon > 0$ and lower the cutoff to $c_\varepsilon = c^S - \varepsilon$.
2. Assign all $v \geq c_\varepsilon$. For assigned negative values, set $u'_\varepsilon(v) = \underline{u}(v)$; otherwise set $u'_\varepsilon(v) = 0$.
3. Participation holds by construction, and promise keeping at $u = 0$ holds because positive tasks generate slack.
4. The new marginal interval has probability $m_\varepsilon = \Pr(c_\varepsilon \leq v < c^S) > 0$.
5. Continuity of Φ gives dynamic payoff at least $\Pi^S + (1 - \delta_p - \delta_p \eta)m_\varepsilon$, which exceeds Π^S for small η .

Thus, when static full assignment is infeasible, the dynamic contract strictly improves on the static optimum.

Proof: Concavity and Symmetry Imply Schur-Concavity

Let Φ be concave and symmetric on a convex symmetric promise set. Fix a Robin Hood transfer from i to j : $\tilde{u} = u - \varepsilon e_i + \varepsilon e_j$, with $u_i > u_j$ and $0 < \varepsilon \leq (u_i - u_j)/2$.

Let σu swap coordinates i and j , and set $\alpha = \varepsilon / (u_i - u_j) \in (0, 1/2]$. Then $\tilde{u} = (1 - \alpha)u + \alpha \sigma u$.

By concavity and symmetry, $\Phi(\tilde{u}) \geq (1 - \alpha)\Phi(u) + \alpha\Phi(\sigma u) = \Phi(u)$.

Thus every Robin Hood transfer weakly raises the value function. Repeating the argument along any finite sequence of transfers gives $u \succeq \tilde{u} \Rightarrow \Phi(\tilde{u}) \geq \Phi(u)$.

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